

Inside one PPO iteration

01 Sample

N parallel worlds $\times$ T control steps; save old log probabilities



02 Estimate

Timeout-aware critic bootstrap $\rightarrow$ GAE advantages and return targets



03 Optimize

E epochs $\times$ M minibatches; clipped policy + robust value loss



04 Diagnose

Raw return, normalized reward, KL, clip fraction, entropy, finiteness



05 Repeat / evaluate

Next rollout uses the new actor; held-out test uses its mean action

A decreasing optimizer loss is not a physical skill test.